



Geobiology

In subject area: [Earth and Planetary Sciences](#)

Geobiology is defined as the study of the interactions between life and the solid Earth, encompassing microbial diversity, metabolism, trace element cycling, and the evolution of life over Earth's history. It is an interdisciplinary field that explores how these interactions have shaped Earth's surface environments and influenced the evolution of life.

AI generated definition based on: [Encyclopedia of Geology \(Second Edition\), 2021](#)

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Chapters and Articles

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Biogeoscience

[2013, Reference Module in Earth Systems and Environmental Sciences](#)

F.S. Peterson

Abstract

Biogeoscience studies cycles of mass and energy between the earth and the life forms that inhabit it. Fundamentally, the field assumes that biology and geology are coupled, and ecosystem functions result from the entangling of physical processes and biological responses. Tools ranging from point counts to long-term sensor networks to remotely-sensed images provide observations, while chemistry and physics provide foundational principles and theory, and statistics are necessary to interpret the results. Because biogeoscience addresses scalar behaviors across multiple scales, robust



methods are necessary to precisely answer scientific inquiry.

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Chapter

Volume 6

2021, *Encyclopedia of Geology (Second Edition)*

Leslie J. Robbins, Kurt O. Konhauser



Abstract

Geobiology, and the subdiscipline of geomicrobiology, are focused on understanding the interactions between life and the solid Earth. This encompasses the entirety of life's interaction with the planet and covers a diverse set of topics including: the diversity of microbial life, the metabolisms of microbial life within the natural environment, the cycling of trace elements, and the evolution of life over Earth's 4-billion-year history. Geomicrobiology, as a subdiscipline, is inherently more focused on the interactions of microbial life with the environment. This is an aspect of geology that is fundamental to our understanding of the Earth system as microbial life has the power to influence a wide variety of processes in both surface and sub-surface environments. In this context, an area of geobiology that has received much focus is how microbes have shaped Earth's surface environments over the last 4.6 billion years and, in turn, how this has affected the evolution of life on Earth. While the field of geobiology continues to grow, having benefited from advances in technology for in-situ measurements, the diversity in topics addressed also continues to expand. In recent years, geobiology has begun to employ traditional geochemical techniques to address medical problems, inform the search for life on Mars and other exoplanets, and provide an important context for assessing future climate scenarios. Given the breadth of topics that geobiology and geomicrobiology seek to address, these are two highly interdisciplinary fields that have led to new insights into how life and the

Earth have interacted in the past and will continue to interact in the future.

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Biosciences: The Study of the Life Sciences

2013, [Reference Module in Earth Systems and Environmental Sciences](#)

M.I. Goldstein

Biosciences

The Biosciences collectively are the life sciences and can be described as the study of the nature, behavior, and uses of living organisms as applied to biology. This section of the Earth Sciences and Environmental Sciences module covers a portion of the Biosciences loosely defined as environmental biology. The Biosciences module is split into the environmental categories of Biogeoscience, Ecology, and Environmental Health.

Biogeoscience

Biogeoscience can best be described as the interactions between biological, chemical and physical processes in the geosphere, hydrosphere and atmosphere.

Biogeoscience includes such studies as the global carbon cycle or nutrient cycles.

Ecology

Ecology can best be described as the relationships of organisms to one another and to their physical surroundings. The broad field of Ecology is narrowed in the Earth Sciences and Environmental Sciences module through a focused set of topic areas, including Community Ecology, Population Ecology, Physiological Ecology, Ecosystem Ecology, Ecological Models, and Ecological Informatics.

Community Ecology, the organization and function of assemblages into communities, includes topics such as how interactions among species and the abiotic environment affect community structure such as patterns of species richness, abundance, and diversity.



Amazing topics such as competition, predation, and mutualism help us understand how assemblages of species give rise to complex assortments of communities.

We also take a look at questions in Population Ecology, factors that affect groups of individuals in a species, and seek answers about how and why populations change over time. Areas of study include population growth, regulation, and dynamics, and demographics.

Physiological Ecology is the study of how organisms adjust to environmental changes. This can occur in the short-term, such as behavioral modifications, or mid-term, such as physiological adjustments. This field looks at how organisms are adapted to their environments and what happens when those environments change, on an individual level, at the population level, and over time. In the long-term, these basic principles are framed in an evolutionary perspective. Evolutionary Ecology addresses the selective pressures on an organism in its environment and the adaptive evolutionary responses to those pressures.

Ecosystems are described in the Ecosystem Ecology section, including the theories and concepts that describe how they function. The biogeochemical cycles may be found in the Biogeoscience section, but Ecosystem Ecology looks deeper into principles of production, decomposition, and the relationships in food webs, including the use of stable isotopes in analysis. Ecosystem Ecology may also look at biological response to stress and disturbance, such as urbanization, fragmentation, fire, or invasive species.

Ecological Models provide insight into how ecological systems work. Basic models provide an avenue to understand systems or components of a system. With the advancement of computational technologies, sophisticated models are increasingly available to provide decision support tools to address management scenarios. In this case, specific predictions maybe made for populations (Population Ecology), communities (Community Ecology), and ecosystems (Ecosystem Ecology). Advanced simulations and mathematical models help address questions with a quantified resolution or error rate. Since Ecological Models are an advanced method to examine a series of assumptions, they are generally iterative, and may be refined over time



when new information is obtained. Ecological Models, when properly used, may offer insight into the pros and cons of certain management decisions, and help evaluate risks. However, acknowledging the limits of the data input into the model is valuable in understanding the reliability of the model output. Whereas a model may help explore a series of assumptions about an Ecological System, the utility of the output should be ground-truthed and verified, where possible.

Ecological Informatics is the cross-disciplinary science of information in the Environmental Sciences. This field of study enables thoughtful and productive management of environmental information, including the integration, analysis, and modelling of ecological, environmental, and social data. We use such information to better understand ecological processes and therefore integrate data within the Earth Sciences and Environmental Sciences with social values and human well-being. Our ability to collect and analyze large datasets has grown significantly in recent years. This is due, in part, to developments in sensor-based data acquisition and internet-based data archiving and sharing, and the use of Ecological Models.



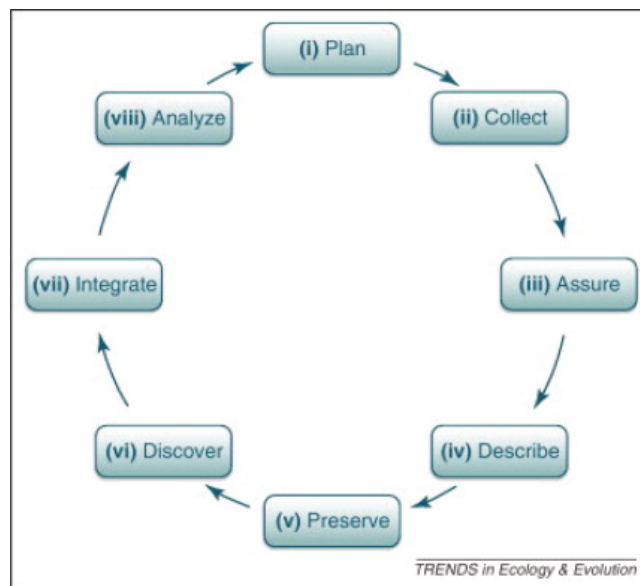


Figure 1. The data life cycle includes the following steps: (i) plan; (ii) collect; (iii) assure (i.e. quality assurance and quality control); (iv) describe (i.e. ascribe metadata); (v) preserve (i.e. deposit data in a secure data repository); (vi) discover (i.e. identify data that might be needed to answer a question); (vii) integrate (e.g. merge data from multiple data sources); and (viii) analyze (e.g. statistical analysis, visualization).

From *Ecoinformatics: supporting ecology as a data-intensive science*. William K. Michener, Matthew B. Jones. *Trends in Ecology & Evolution*, Volume 27, Issue 2, 85–93, 11 January 2012. Copyright © 2011 Elsevier Ltd All rights reserved. 10.1016/j.tree.2011.11.016.

<http://download.cell.com/images/journalimages/0169-5347/PIIS0169534711003399.gr1.lrg.jpg>

Environmental Health

Environmental Health is the broad topic addressing human health and disease and the factors in the environment that can potentially affect health. The Biosciences section of the Earth Sciences and Environmental Sciences module contains topics addressing Globalization, Environmental Biotechnology, Environmental Toxicology, Exposure, Risk Assessment, and Children's Health.

Globalization of the environment can be loosely defined as the increasingly global effects of human activity on the environment. Many positive benefits have resulted from increased global economic integration, and much concern has been discussed on the important conceptual change to viewing environmental protections as an international concern, particularly due to cross-border or trans-boundary effects of environmental change.

Environmental Biotechnology integrates science and engineering to develop new approaches for sustainable development. These technologies employ the potential

uses of microorganisms, plants, and animals for biological remediation, restoration, new manufacturing technologies, and sustainable development.

Environmental Toxicology studies contaminants in the environment, including cause and effect relationships between contaminants and organisms, lethal and sub-lethal exposure scenarios, and experimental studies of potential scenarios of Exposure. Risk Assessments can be derived when information on routes of Exposure is used in combination with results from studies in Environmental Toxicology.

Environmental Health is important for mental and physical development of children. Children's Health issues, such as asthma, lead exposure, cancer, diabetes, and developmental disabilities, have been linked to environmental exposures.

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Geologic History and Energy

2018, [Encyclopedia of the Anthropocene](#)

S.A. Elias

Bioscience

The aspects of the biological sciences covered in the module include Biogeoscience, Ecology, and Environmental Health. Biogeoscience is the study of interactions of the biosphere with the lithosphere, hydrosphere and atmosphere. This is a rapidly developing field, reflecting the fact that scientists are becoming increasingly aware of these interactions between the various components of the global system. Ecology is a vast research area than began in earnest in the early 20th century and has expanded ever since. It concerns the interactions between organisms and their environments, at various spatial scales ranging from local Ecological Communities to global Ecosystems. Environmental pressures affect organisms at the



individual level, as discussed in the articles on Physiological Ecology and Evolutionary Ecology. The effects of environmental change are also felt at the population level, as discussed in the [Population Ecology](#) articles. One of the chief methods of ecological analyses is through computer modelling, as discussed in the Ecological Informatics and Ecological Models sections. The study of how ecosystems function is far from an esoteric exercise. We must gain as great an understanding of these principles as possible, if we hope to maintain the remaining fragments of the natural world. Regions containing undisturbed, natural habitats are increasingly being taken for various human uses, such as agriculture and industry, towns and cities, and the roads connecting them. The fragmentation of natural habitats is one of the most damaging of [human impacts](#) on the natural world, playing a central role in the modern extinction crisis discussed above (Dobson et al., 2006).

The section on Environmental health mainly deals with Human Health, although the articles are generally framed in the context that the health of the planet as a whole has direct effects on the health of our species. Sadly, we humans are the primary agents in the degradation of our world's environment. These impacts can be blatantly obvious, such as Bioterrorism, [Chemical Warfare](#), and exposure to man-made toxic chemicals (covered in [Environmental Health Exposure](#) and Environmental Health Toxicology and Systems Toxicology). But they can be more subtle and harder to link directly to human agencies, such as the [effects of Climate Change](#), [Genetically Modified Organisms](#), and Globalization of trade. In a worst-case scenario, Environmental Health Disasters kill hundreds or thousands of people, such as the poisoning of more than 15000 people from the mass-release of pesticide from the Union Carbide factory in Bhopal, India in 1984 (Jasanoff, 1994) or the exposure of many thousands of people to radiation following the nuclear accident at Chernobyl, Ukraine in 1986 (Balonov and Bouville, 2011). Issues of Environmental Health Justice and the Ethics of [environmental issues](#) are spawned by both large-scale disasters and day-by-day environmental degradation. These are almost always difficult issues to tackle in the modern world, where multinational corporations operate around the world, so that the [Environmental Health](#)



Policy and Agencies of a given country have little or no authority over the overseas activities of large corporations. Regional Environmental Health issues must take into account the Physical and Social Environment, and the Economics of Environmental Health inevitably play a part in decision making, for better or worse. Advances in Environmental Health Biotechnology and Environmental Health Medicine can ameliorate environmental health problems to a certain extent, but ultimately the environmental health of any nation must be safeguarded by the leaders of industry and government, working together for the common good.

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Chapter

Biosignatures—The prime targets in the search for life beyond Earth

2023, [Water Worlds in the Solar System](#)

Antony Joseph

4.7 Identification of biosignature in Antarctic rocks

Deciphering biosignatures and evidence of microbial activity in ancient rock remains a central challenge in geobiology studies (e.g., Rosing, 1999; Fedo and Whitehouse, 2002; Lepland et al., 2005). When surface-derived, organic-bearing rocks are transferred to Earth's shallow interior, the combination of burial and deformation can ultimately make it impossible to distinguish a biological signature in relict carbonaceous compounds (Brasier et al., 2005). Consequently, determination of the degree of metamorphism beyond which life's signatures are no longer recognizable in ancient carbon is a research topic of considerable interest in early Earth and extra-terrestrial studies (e.g., Schopf and Kudryavtsev, 2009; Glikson et al., 2008; Oehler et al., 2009). For example, a recent approach focuses on the applicability and limitations of using Raman



spectroscopy to characterize evidence of ancient life (e.g., van Zuilen et al., 2002; Schopf et al., 2005; McKeegan et al., 2007; Schiffbauer et al., 2007; van Zuilen et al., 2007; Marshall et al., 2007).

The cold, dry ecosystems of Antarctica are known to harbor traces left behind by microbial activity within certain types of rocks. Some indirect biomarkers of cryptoendolithic activity (i.e., ability of microorganisms to colonize the empty spaces or pores inside a rock) in the Antarctic cold desert zone have been described in the literature; primarily geophysical and geochemical bioweathering patterns macroscopically observed in sandstone rock. Wierzchos et al. (2003) have shown that in this extreme environment, minerals are biologically transformed, and as a result, Fe-rich diagenetic minerals (i.e., recombination or rearrangement of chemical or mineral constituents, resulting in a new product) in the form of iron hydroxide nanocrystals and biogenic clays are deposited around chasmoendolithic hyphae (long, branching filamentous structured organisms, growing in the interior of rocks, inhabiting fissures and cracks of the rock) and bacterial cells. Thus, when microbial life decays, these characteristic neocrystallized minerals act as distinct biomarkers of previous endolithic activity. Wierzchos et al. (2003) have suggested that the ability to recognize these traces may have potential astrobiological implications because the Antarctic Ross Desert is considered a terrestrial analogue of a possible ecosystem on early Mars.

The cryptoendolithic habitat of the Antarctic Dry Valleys has been considered a good analogy for past Martian ecosystems, if life arose on the planet. Yet cryptoendoliths are thought to favor the colonization of rocks that have a pre-existing porous structure, for example, sandstones. This may weaken their significance as exact analogues of potential rock-colonizing organisms on Mars, given our current understanding of the dominant volcanic nature of Martian geology. However, the production of oxalic acid, by these lichen-dominated communities, and its weathering potential indicate that it could be an aid in rock colonization, enabling endoliths (organisms that live inside rocks or in pores between mineral grains) to inhabit a wider variety of rock types. Utilizing ICP-AES and scanning electron microscope techniques, a study carried out by Blackhurst et al. (2005) investigated elemental and mineralogical



compositions within colonized and uncolonized layers in individual sandstone samples. This was in order to determine if the weathering of mineral phases within the colonized layers causes an increase in the amount of pore space available for colonization. The results show that colonized layers are more weathered than uncolonized, deeper portions of the rock substrate. It was found that layers within uncolonized samples have uniform compositions. Differences between the colonized and uncolonized layers also occur to varying extents within colonized rocks of different mineralogical maturities. The results confirm that cryptoendoliths modify their habitat through the production of oxalic acid and suggest that over time this directly increases the porosity of their inhabited layer, potentially increasing the biomass it can support (Blackhurst et al., 2005).



In terms of possible detection of signs of extra-terrestrial life, a meteorite that reached Earth from Mars (the meteorite was found to have been encased in Antarctic ice for 13,000 years and lay exposed on the surface for ~500 years (McKay et al., 1996)) attracted considerable interest and attention not only among planetary scientists and astrobiologists but also among administrators and the general public. For example, quoting from Vago et al. (2017), “In 1996, David McKay and his colleagues published the first description of possible microbial signatures in extraterrestrial rocks, namely in a meteorite from Mars called ALH84001 (McKay et al., 1996). The subject was so delicate that President Bill Clinton announced the news in a press conference (Statement, 1996). The ensuing interest in the scientific world spurred a huge increase in astrobiological research and, in particular, the study of biosignatures.”

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Chapter

Working in the in-between: Defining the boundary layer

2023, [Conceptual Boundary Layer Meteorology](#)

1.5 Beyond the air

As previously mentioned, boundary layer meteorologists come from a variety of disciplines. Many now work in the area of biogeosciences. This chapter briefly discussed the water and carbon cycles as an introduction to the larger group of cycles known as biogeochemical cycles or the flow of gasses and their chemical changes through the earth–atmosphere system. While water and carbon are the two most important, other biogeochemical cycles are also of interest. These include non-carbon gasses such as nitrous oxide (N₂O), ammonium (NH₃), mercury (Hg), and others. Some have coined the term eco-climatology to represent the work being done in this area. While water, carbon, and nitrogen are extremely important in the boundary layer we have chosen not to include chapters on them explicitly; for the interested reader I recommend Bonan (2016).

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Biogeochemistry

2014, *Treatise on Geochemistry (Second Edition)*

David M. Karl, William H. Schlesinger

What's in a name ... everything under the sun, especially when the name is *Biogeochemistry*. Biogeochemistry is the term used to describe one of the most important subdisciplines of *Biogeosciences*, the holistic study of the Earth system. Biogeochemical research includes theoretical, observational, experimental, and modeling components, and now spans from molecular studies to the largest scales in space and time that affect our planet. Therefore, as a scientific discipline, biogeochemistry is both integrative and transdisciplinary, and it is attracting a new and diverse spectrum of investigators each year. Many of the more successful field-oriented campaigns involve collaborations among individuals who do not normally interact, where the whole of the coordinated effort is much greater than the sum of its parts.



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Chapter

Co-Evolution of Life and Plate Tectonics: The Biogeodynamic Perspective on the Mesoproterozoic-Neoproterozoic Transitions

2023, [Dynamics of Plate Tectonics and Mantle Convection](#)

Robert J. Stern, Taras V. Gerya



Abstract

There is growing understanding that life evolution on Earth is intimately related to and shaped by the global tectono-magmatic style that evolved through geological time until modern plate tectonics was established in the Neoproterozoic, between 1000 and 541 million years ago. The co-evolution of life and tectonics on Earth is the subject of Biogeodynamics—an emerging field at the interface between the disciplines of geodynamics, geomorphology, climate, ocean and atmosphere sciences, geobiology, and ecology. Biogeodynamics focuses on how evolution of the planetary interiors, surface, atmosphere, ocean, climate, and life interact and interrelate. Here, we review the state of the art for our understanding of co-evolution of modern plate tectonics and complex life. We start by explaining what Biogeodynamics is, then explain what are the modern plate tectonics and single-lid tectonics and how these differ, and then discuss how plate tectonics controls global terrestrial biodiversity today. We summarize first biodiversity in the Mesoproterozoic and Neoproterozoic, including how evolution accelerated remarkably in the Neoproterozoic and then summarize geological evidence for a protracted Mesoproterozoic single lid that was slowly transformed in Neoproterozoic time into the modern global plate tectonic mosaic. We next explore how the Neoproterozoic tectonic revolution enhanced five factors that accelerated biological evolution: (1) increased nutrient supply; (2) increased free oxygen in the

atmosphere and ocean; (3) climate amelioration; (4) accelerated habitat formation; and (5) moderate sustained environmental pressure. Finally, we suggest some avenues for future biogeodynamic research into the co-evolution of Earth's tectonics and Biosphere. A 5-min video overview can be seen at <https://www.youtube.com/watch?v=4jaMCz71ejc&t=5s>.

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Chapter

An integrated approach to landscape evolution

2021, [Landscape Evolution](#)

Jonathan D. Phillips

Comparison and contrast: integrated approach to landscape evolution versus traditional approaches

The integrated approach to landscape evolution described here is not totally novel, as key elements already exist. However, there are certain characteristics that distinguish it from existing theories, concepts, and models. These are outlined below.

It is safe to say that no geologist, ecologist, soil scientist, geographer, etc., ever truly believed that the systems they were focused on developed independently of other environmental factors and systems. It is also safe to say that they (we) dealt with the “other” systems only to the extent necessary to understand the system of interest. Thus, notions of evolution of geomorphic landscapes, ecosystems, and soils developed along lines that acknowledged and even incorporated the effects of the other systems, but that were generally unconcerned with reciprocal interactions and coevolution. For example, much research in applied climatology and hydrology up to the 21st century was oriented toward the water use and needs of plants, but the hydroclimatological effects of vegetation were mostly limited to consideration of the transpiration “pump.” Likewise, soil geomorphology explored the influences of topography and



geomorphology on soils, but mostly treated effects of soil properties on geomorphic processes very simplistically. The situation has changed somewhat with the rise of biogeomorphology, ecohydrology, geobiology, critical zone science, etc., but there has yet to arise a theory of coevolution of ecological, geomorphic, and soil systems that treats all of these with comparable levels of detail and sophistication.

Beyond the fact that these ESS do indeed coevolve, another incentive for an integrated coevolutionary approach is the potential for the subdisciplines of pedology, ecology, geomorphology, and hydrology to inform each other. For example, the factorial approach developed by pedologists has been adapted to studies of ecology, geomorphology, Quaternary geology, palaeoenvironmental reconstruction, and archeology. Other examples could be cited: the energy, mass, and nutrient budget approach first embraced by ecologists is now firmly entrenched in geomorphology and geochemistry, and approaches to linking form or pattern and process pioneered in geomorphology are part of the backbone of landscape ecology. We have no doubt only scratched the surface of these possibilities.

Many of the classic theories outlined in Background: evolutionary pathways in landscapes are single-path, single-outcome types—Clementsian succession, the cycle of erosion, progress toward normal or zonal soils, etc. Equilibrium-based theories, such as dynamic equilibrium in geomorphology and steady-state regolith thickness, are also single-outcome concepts, as are suggestions, common in the 1990s and early 2000s, that geomorphic (and other) systems evolve toward a self-organized critical state (see Hergarten's review) (Hergarten, 2002).

There also exist a number of existing theories, however, that are not based on progress toward a single attractor state, be the latter a zonal soil, peneplain, steady-state, climax community or critical state. These include assembly theories and alternative stable state concepts in ecology, “nonequilibrium” theories in geomorphology, and multipath concepts of pedogenesis.

The approach here recognizes what by now must be considered the obvious reality of multiple pathways and outcomes in ESS, and is thus aligned with the tradition of concepts that embrace those possibilities. However, there



is no effort to reject or deny the possibility of, for instance, progressive downwearing toward an erosion surface, development of zonal soils, or succession toward a climate-controlled climax. Rather, these are seen as particular cases out of multiple possibilities. Likewise, I do not deny the existence of steady-state equilibria (or their utility as a reference condition or model assumption)—I just do not treat these as normative; as necessarily more common, natural, important, or desirable than unstable or nonequilibrium conditions.

Neither do I view landscape evolution as an unrestricted, anything-is-possible proposition. Invariant laws of nature, and physical constraints of a given environment, make some pathways and outcomes impossible or improbable, and others possible or likely. The multiple pathways and outcomes occur within a defined evolution space, as described by (Phillips, 2018c).

The integrated coevolutionary approach espoused here is based on a notion of incorporating contingency—place and history. Moreover, I seek to put geographical and historical contingency on an equal footing with laws in explaining landscape evolution. For particular problems, either laws, place, or history may be paramount as a determining factor. However, in the aggregate, all are of comparable importance.

The notion of selection in a broad sense as described earlier is another key component of the approach. Landscape evolution is seen as guided by (probabilistic) selection that sees to the preferential formation, persistence, and propagation of the most efficient and/or durable forms, features, and patterns.

My approach to understanding ESSs and the evolution of landscapes is, for better or worse, firmly based on a view of the world as a set of highly interconnected components, and that a network-type model is the best way to represent this. One advantage of this is that network representation lends itself to a variety of analytical techniques, including nonlinear dynamical systems theory, graph theory, and others.

The need or demand for integration of landscape evolution theories comes from experience—my own and the scientific community in general. Geomorphologists, ecologists, and hydrologists have long recognized, for example, that one cannot really understand any aspect of



wetlands without considering the interaction of hydrological, geomorphological, and ecological processes. This need came into even sharper focus in the study of problems such as coastal wetland responses to sea-level rise, where interactions and feedbacks are now a standard part of both models and field studies as described in the example above. This kind of interconnectivity may be more clearly evident in such highly dynamic environments or in those undergoing rapid or recent change. However, the eco-geo-hydro-pedo interactions are equally important in virtually all ESS.

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Chapter

Earth surface systems as supraorganisms

2021, [Landscape Evolution](#)

Jonathan D. Phillips

Background

The interconnectedness of biota, geophysical and geochemical phenomena, landforms, and soils has long been recognized. Milestones include Dokuchaev's (1883) pioneering ideas of soils as the product of the combined influences of geology, climate, topography, and organisms, Vernadsky's (1926) biosphere concept and foundational work in biogeochemistry, and the ecosystem idea (Tansley, 1935). Dokuchaev (1883) did not explicitly deal with reciprocal interactions, but did recognize that the factors of soil formation influence, and are influenced by, the other factors as well as the soil itself. The idea of ecosystems (and possibly the term) predates Tansley, but his 1935 article introduced the concept to a broad audience.

Until relatively recently, however, studies tended to be one-way in terms of causality—for example, the effects of climate on geomorphic processes or the influence of topography on soil or vegetation distributions—rather than reciprocal interactions, as described in Chapter 1.



The increasing focus on reciprocity, along with the recognition that biota, soils, landforms, hydrology, and climate are best understood as responding and evolving together led to the rise of several subdisciplines focusing on these reciprocal interactions. These include *biogeomorphology* (or *ecogeomorphology*), which is “concerned with the influence of landforms on the distribution and development of plants, animals and microorganisms; and with the influence of plants, animals and microorganisms on earth surface processes and the development of landforms” (Viles, 1988). *Ecohydrology* is at the interface of hydrology and ecology, and according to the mission statement for the journal of that name, emphasizes interactions and associated feedbacks in both space and time between ecological systems and the hydrological cycle. The journal *Geobiology* describes that subfield as exploring the relationship between life and the Earth's physical and chemical environment. *Hydropedology* is portrayed as bridging traditional pedology with soil physics and hydrology (Lin, 2010; 2006; Kutilek and Nielsen, 2007).

Recent decades have also seen the rise of the terms *Earth system*, *climate*, *ecosystem*, and *critical zone science*. Earth system science explicitly recognizes the dense interconnections of the atmo-, hydro-, litho-, and biospheres, and is particularly concerned with addressing these at very broad—including global—scales and a range of temporal scales, from fluid dynamics to Earth evolution. The term *climate science* seems to have emerged as an attempt to encompass climatology, paleoclimatology, and atmospheric sciences, and also the study of climate impacts on, and feedbacks with, human and other biophysical systems. *Ecosystem science* emphasizes that the field transcends ecology. According to the mission statement of the journal *Ecosystems*, “the scope of ecosystem science extends from bounded systems such as watersheds to spatially complex landscapes, to the Earth itself, and crosses temporal scales from seconds to millennia. Ecosystem science has strong links to other disciplines including landscape ecology, global ecology, biogeochemistry, aquatic ecology, soil science, hydrology, ecological economics and conservation biology.” In the US National Science Foundation Critical Zone program, the critical zone is defined as “Earth's permeable near-surface layer... from the tops of the trees to the bottom of the groundwater”



(<http://criticalzone.org/national/research/the-critical-zone-1national/>). The critical zone (the term is in wide use outside the USA, with similar definitions) is an integrated approach to the study of rock, regolith, soil, water, biota, and atmosphere interactions near Earth's surface.

The recent emergence of these subdisciplines, terms, and concepts (along with older ones such as biogeochemistry and geocology) is clear evidence of recognition that (1) no component of Earth's system(s) can be fully understood in isolation; (2) ecosystems and other Earth surface systems are characterized by constant internal and external feedbacks and reciprocal interactions. This does not necessarily equate to supraorganic notions, but is consistent with a view of ecosystems as supraorganisms.

Explicitly supraorganic concepts of ecosystems already mentioned above include Rowe (2001), Jackson (2010), and Brunetti (2014). In addition, several other authors have argued that ecological systems can evolve—at the system level—because of selection processes. Smith's (1986) overlooked biotic/abiotic conceptual model holds that adaptations are the structural attributes that mediate energy dissipation. Thus “it is the properties of spatial interaction that evolve, not the organisms themselves.” Responses to and influences on the abiotic environment within the biotic sector are guided by selection for more efficient means of turning over material resources (Smith, 1986, p. 250). Similarly, biogeochemical selection—that is, selection for advantages associated with faster energy and nutrient (re)cycling—can account for “Gaian” supraorganic evolution, Lapenis (2002) showed. He linked these ideas to ecological, evolutionary, and biogeochemical theories that emerged in Russia in the early 20th century from Vernadsky, R.V. Rizpolozhensky, P.A. Kropotkin, and V.A. Kostitzin. Lekevicius (2006) has called a view that ecosystems (as opposed to individuals, species, genes, or any other division) are the fundamental unit of life the “Russian Paradigm” and traces its roots to Vernadsky and S.N. Winogradsky. As described by Lekevicius (2006), this paradigm is entirely consistent with a supraorganic view of ecosystems. Savenjie and Hrachowitz (2017) described hydrological catchments as metaorganisms, where the most important active agents are ecosystems. These evolve so as to increase the efficiency of infiltration and



retention of moisture, mobilization, and retention of nutrients, and drainage, mainly through the root zone. Hydrological systems are alive and have the capacity to adjust to environmental conditions, Savenjie and Hrachowitz (2017) maintained, via selection processes.

A complete review of such concepts is beyond the scope of this chapter, but several additional threads are reviewed by Lapenis (2002), Phillips (2008), and Lekevicius (2002). I return to evolution at the ecosystem level later in this chapter.

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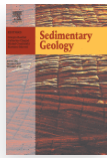
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